Object-Oriented Software Engineering

Using UML, Patterns, and Java



Outline of this Class

- What is UML?
- A more detailed view on
  - ✓ Use case diagrams
  - ✓ Class diagrams
  - ✓ Sequence diagrams
  - ➢ Activity diagrams

What is UML? Unified Modeling Language

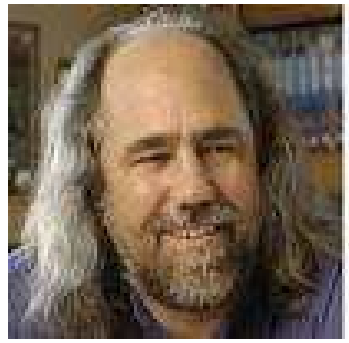
- Convergence of different notations used in object-oriented methods, mainly
  - OMT (James Rumbaugh and collegues), OOSE (Ivar Jacobson), Booch (Grady Booch)
- They also developed the Rational Unified Process, which became the Unified Process in 1999



25 year at GE Research, where he developed OMT, joined (IBM) Rational in 1994, CASE tool OMTool



At Ericsson until 1994, developed use cases and the CASE tool Objectory, at IBM Rational since 1995, http://www.ivarjacobson.com



Developed the Booch method ("clouds"), ACM Fellow 1995, and IBM Fellow 2003 http://www.booch.com/

UML

- Nonproprietary standard for modeling systems
- Current Version: UML 2.2
  - Information at the OMG portal http://www.uml.org/
- Commercial tools:
  - Rational (IBM),Together (Borland), Visual Architect (Visual Paradigm), Enterprise Architect (Sparx Systems)
- Open Source tools http://www.sourceforge.net/
  - ArgoUML, StarUML, Umbrello (for KDE), PoseidonUML
- Example of research tools: Unicase, Sysiphus
  - Based on a unified project model for modeling, collaboration and project organization
  - http://unicase.org
  - http://sysiphus.in.tum.de/



UML Basic Notation: First Summary

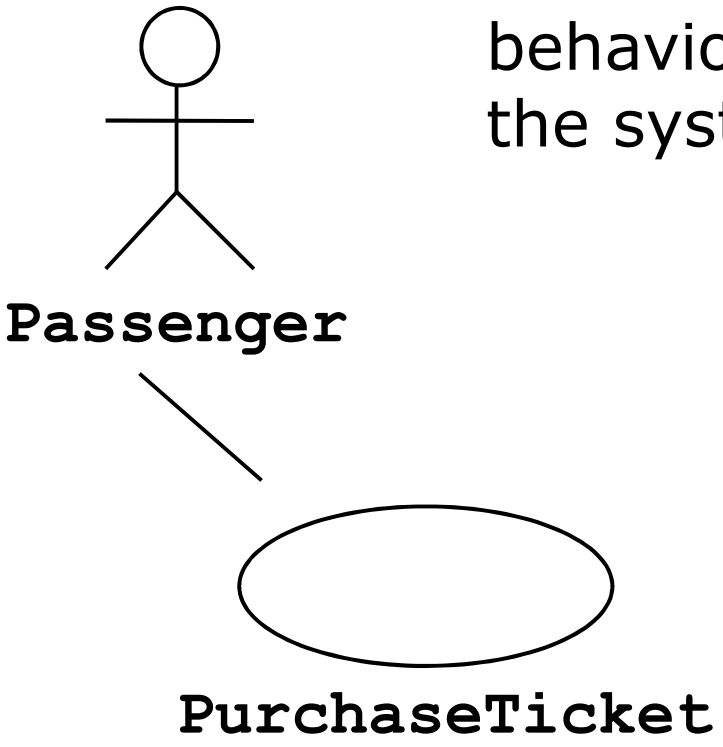
- UML provides a wide variety of notations for modeling many aspects of software systems
- In the last lecture did a first pass on:
  - Functional model: Use case diagrams
  - Object model: Class diagrams
  - Dynamic model: Sequence diagrams, statechart diagram

- Now we go into a little bit more detail…

UML First Pass (covered in Last Lecture)

- Use case diagrams
  - Describe the functional behavior of the system as seen by the user
- Class diagrams
  - Describe the static structure of the system: Objects, attributes, associations
- Sequence diagrams
  - Describe the dynamic behavior between objects of the system
- Statechart diagrams
  - Describe the dynamic behavior of an individual object
- Activity diagrams
  - Describe the dynamic behavior of a system, in particular the workflow.

UML Use Case Diagrams



Used during requirements elicitation and analysis to represent external behavior ("visible from the outside of the system")

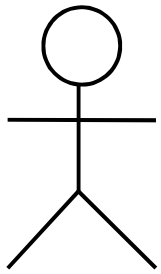
An ***Actor*** represents a role, that is, a type of user of the system

A ***use case*** represents a class of functionality provided by the system

Use case model:

The set of all use cases that completely describe the functionality of the system.

Actors



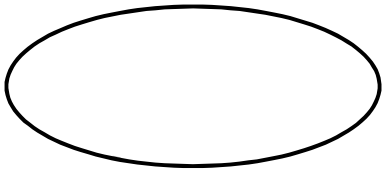
Passenger

- An actor is a model for an external entity which interacts (communicates) with the system:
  - User
  - External system (Another system)
  - Physical environment (e.g. Weather)
- An actor has a unique name and an optional description
- Examples:
  - **Passenger**: A person in the train
  - **GPS satellite**: An external system that provides the system with GPS coordinates.

Optional Description

Name

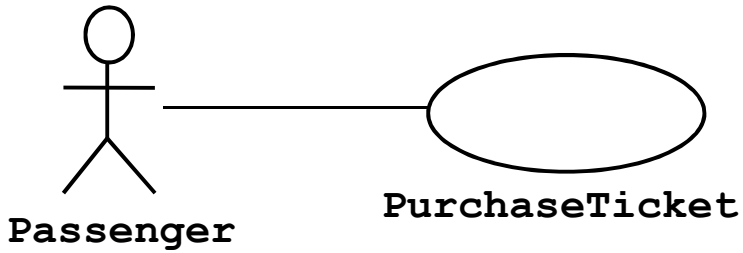
Use Case



PurchaseTicket

- A use case represents a class of functionality provided by the system
- Use cases can be described textually, with a focus on the event flow between actor and system
- The textual use case description consists of 6 parts:
  1. Unique name
  2. Participating actors
  3. Entry conditions
  4. Exit conditions
  5. Flow of events
  6. Special requirements.

Textual Use Case Description Example



1. Name: `Purchase ticket`

2. Participating actor:
`Passenger`

3. Entry condition:

- `Passenger` stands in front of ticket distributor
- `Passenger` has sufficient money to purchase ticket

4. Exit condition:

- `Passenger` has ticket

5. Flow of events:

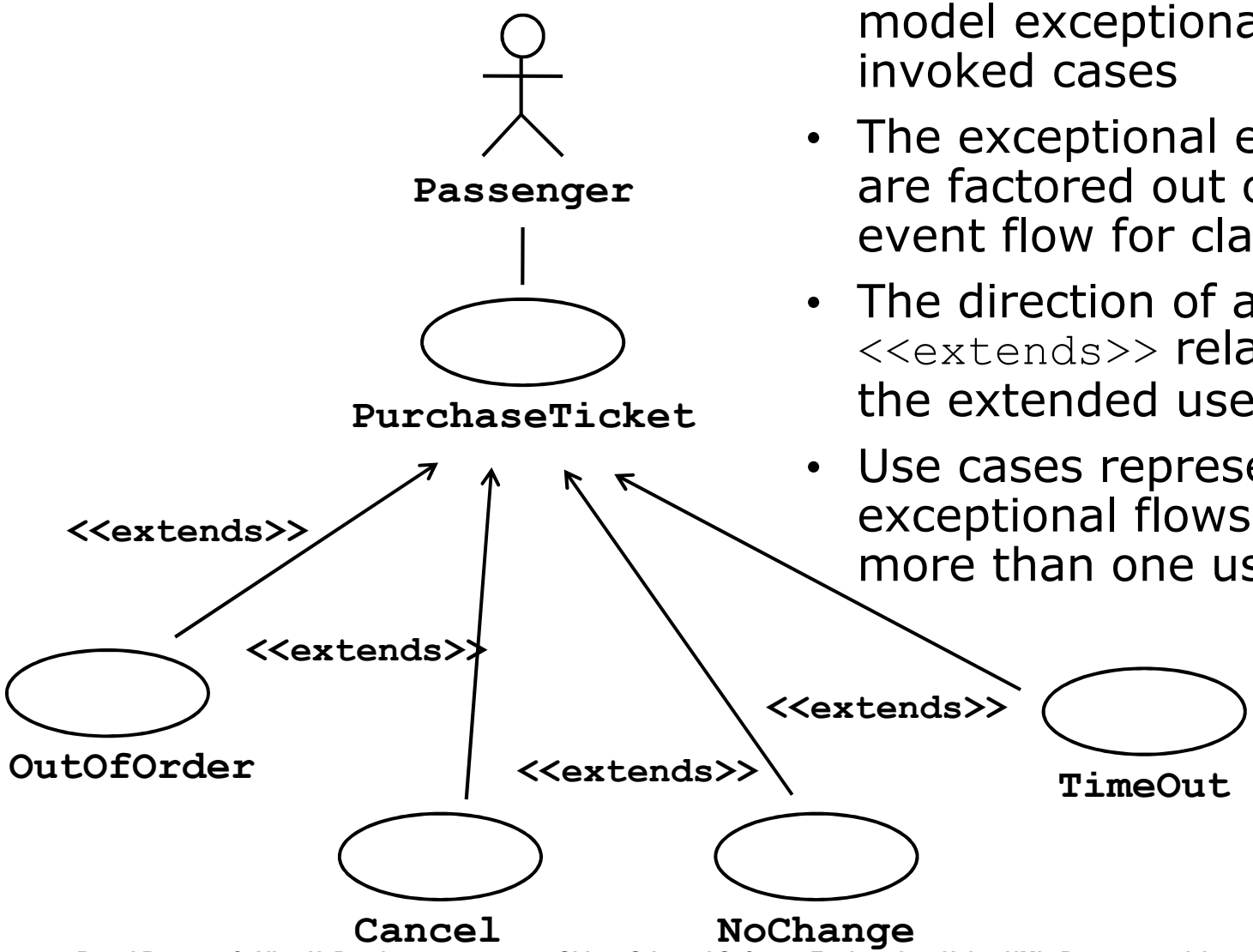
1. `Passenger` selects the number of zones to be traveled
2. Ticket Distributor displays the amount due
3. `Passenger` inserts money, at least the amount due
4. Ticket Distributor returns change
5. Ticket Distributor issues ticket

6. Special requirements:
None.

Uses Cases can be related

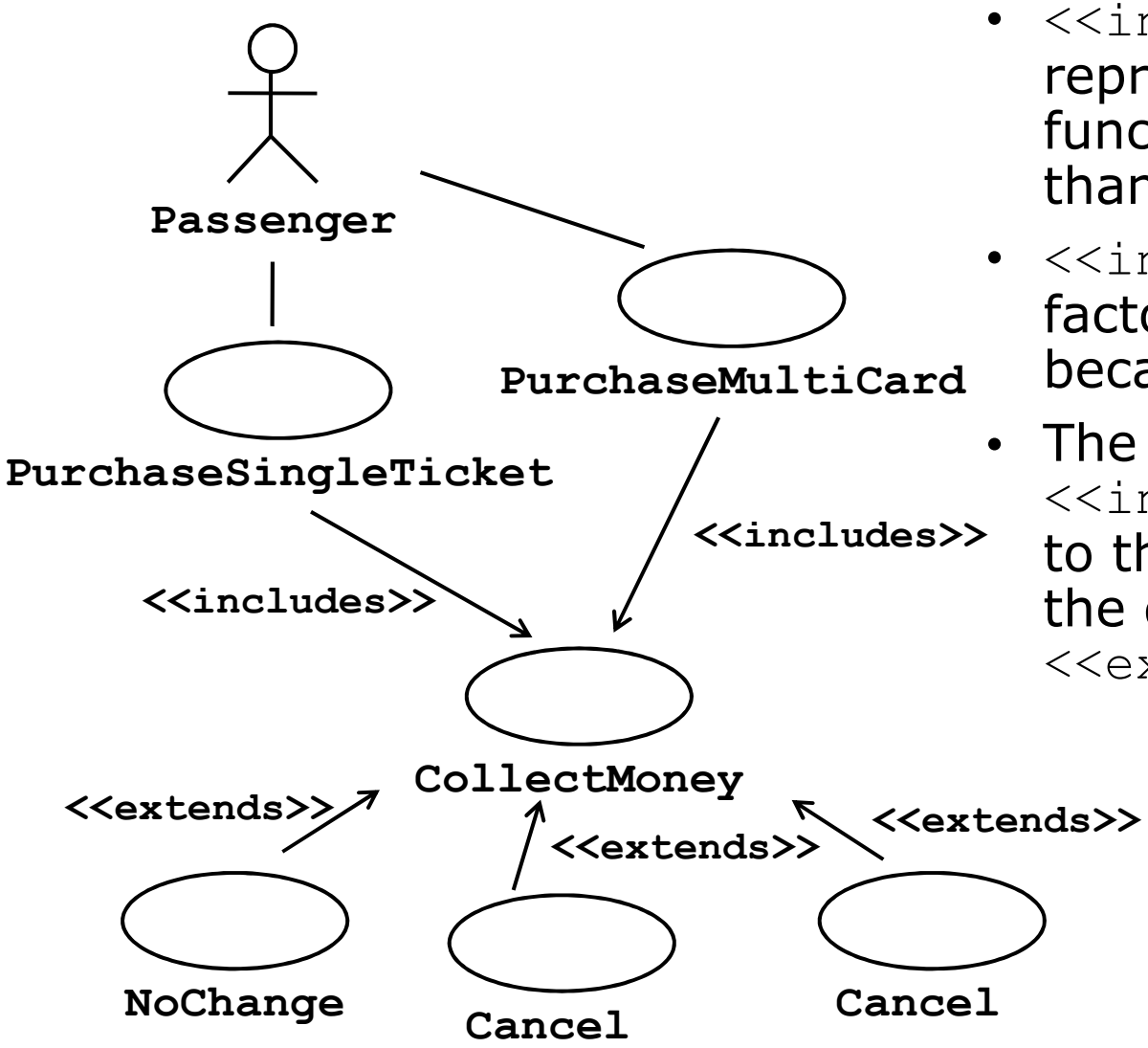
- Extends Relationship
  - To represent seldom invoked use cases or exceptional functionality
- Includes Relationship
  - To represent functional behavior common to more than one use case.

The <<*extends*>> Relationship



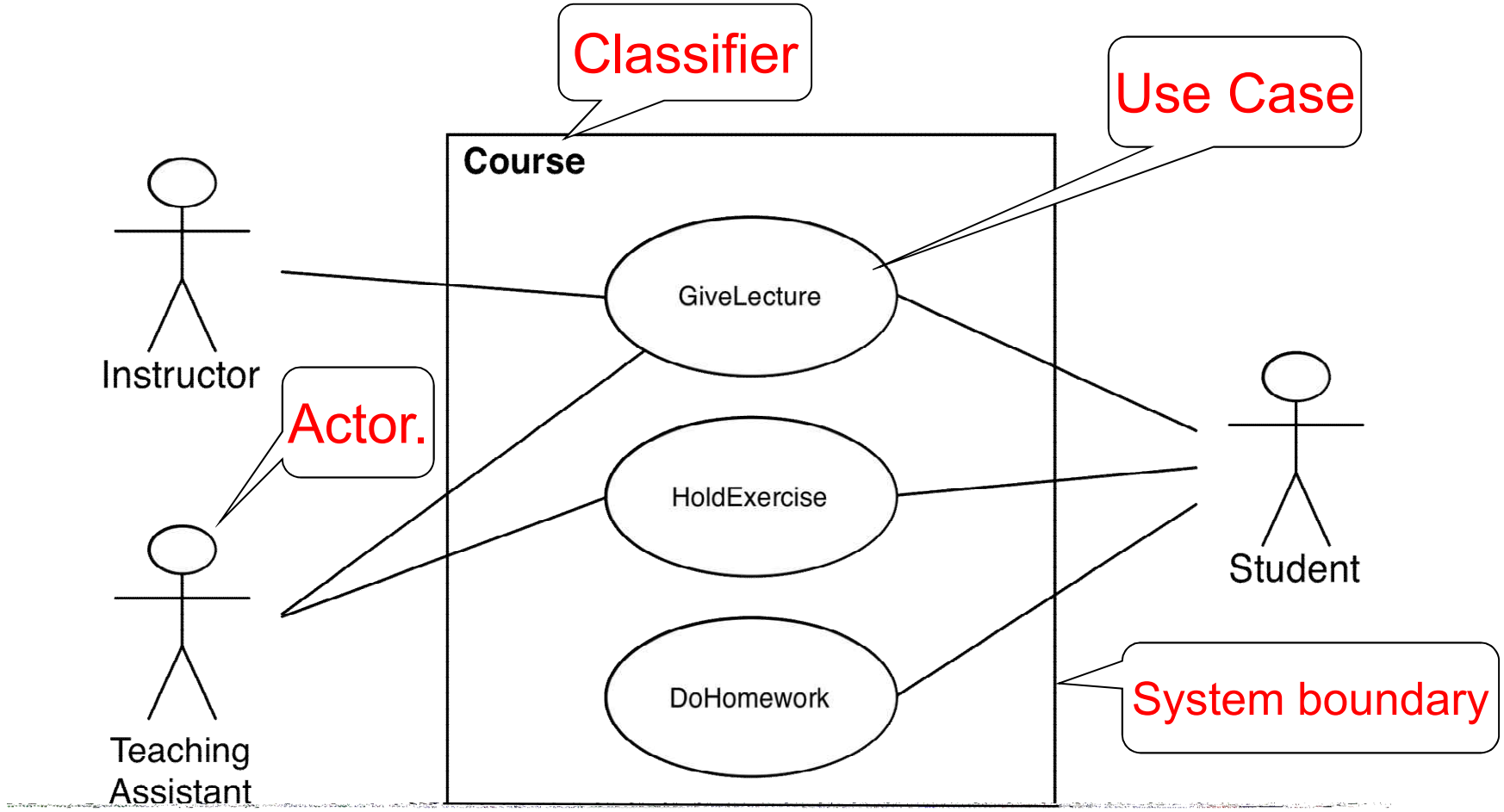
- `<<extends>>` relationships model exceptional or seldom invoked cases
- The exceptional event flows are factored out of the main event flow for clarity
- The direction of an `<<extends>>` relationship is to the extended use case
- Use cases representing exceptional flows can extend more than one use case.

The *<<includes>>* Relationship

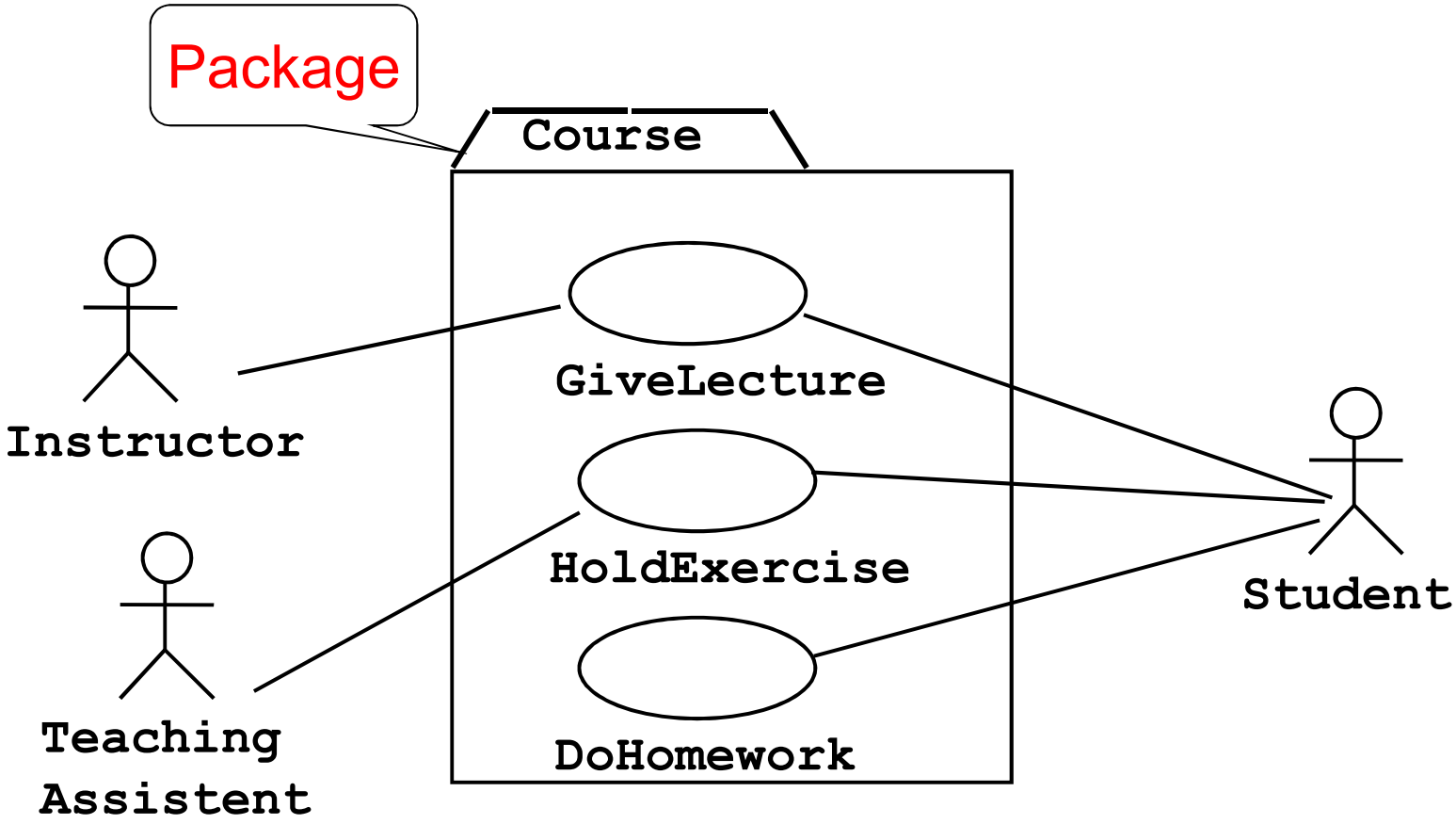


- <<includes>> relationship represents common functionality needed in more than one use case
- <<includes>> behavior is factored out for reuse, not because it is an exception
- The direction of a <<includes>> relationship is to the using use case (unlike the direction of the <<extends>> relationship).

Use Case Models can be packaged

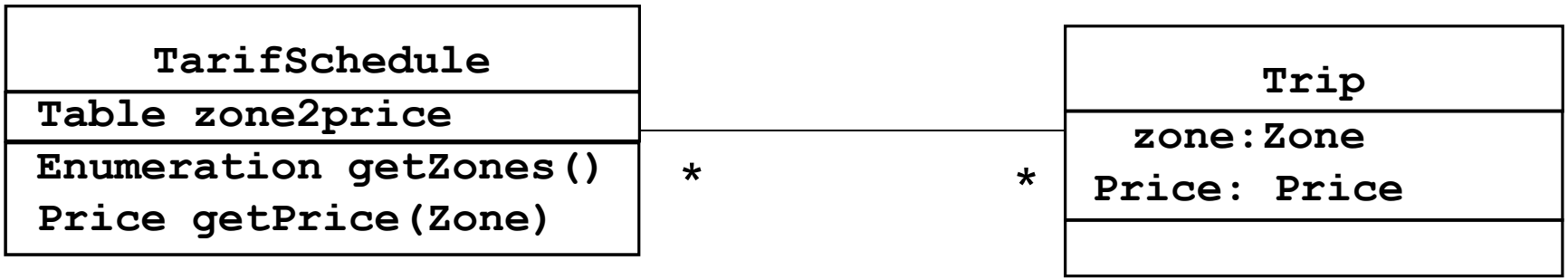


Historical Remark: UML 1 used packages

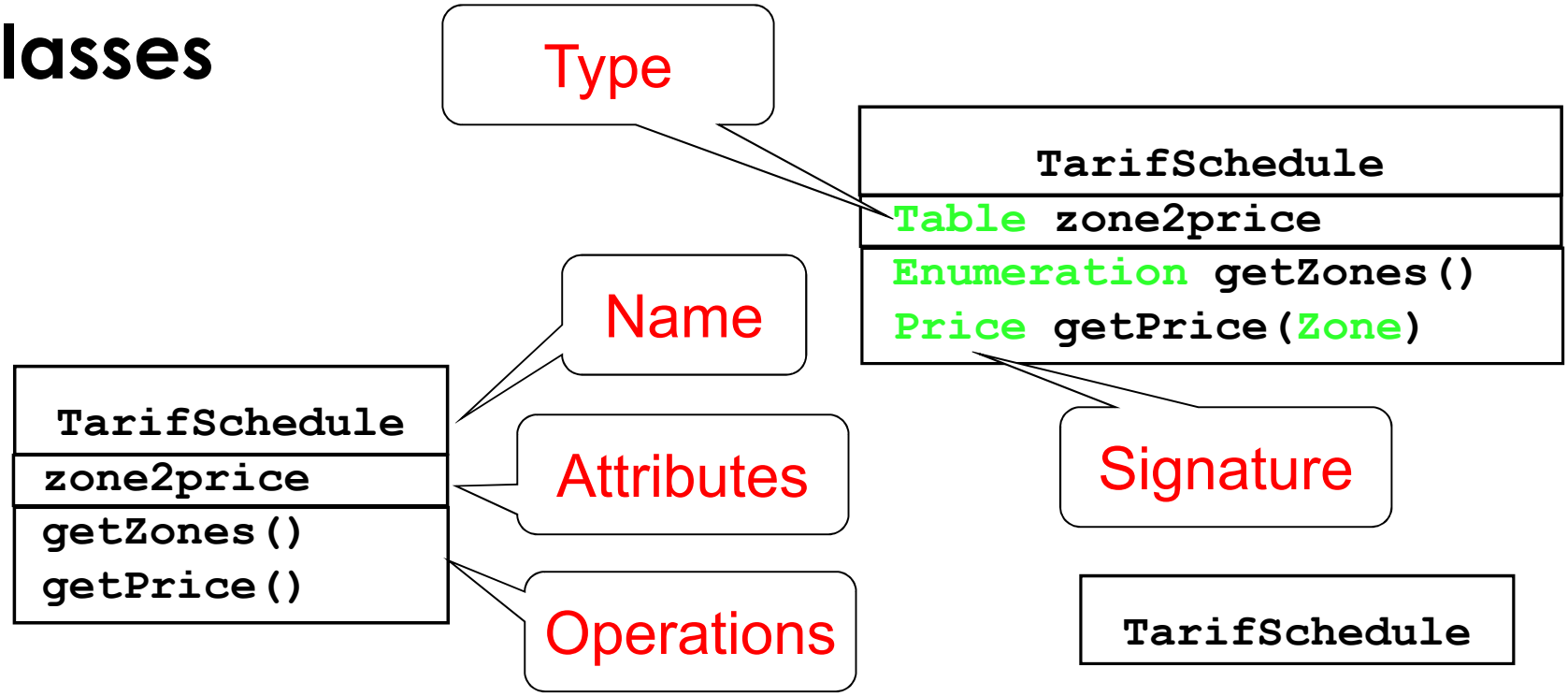


Class Diagrams

- Class diagrams represent the structure of the system
- Used
  - during requirements analysis to model application domain concepts
  - during system design to model subsystems
  - during object design to specify the detailed behavior and attributes of classes.



Classes



- A ***class*** represents a concept
- A class encapsulates state ***(attributes)*** and behavior ***(operations)***

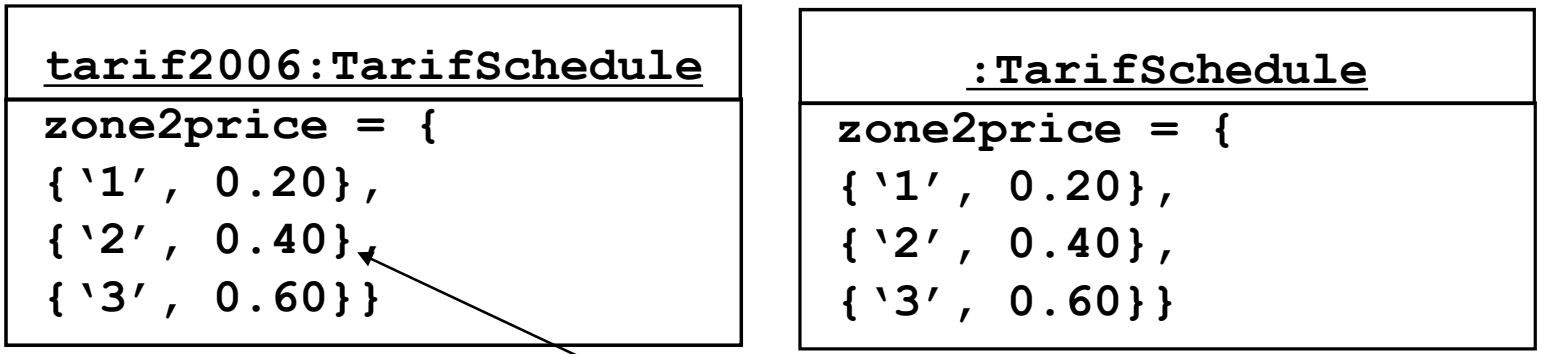
  Each attribute has a ***type***
  Each operation has a ***signature***

The class name is the only mandatory information

Actor vs Class vs Object

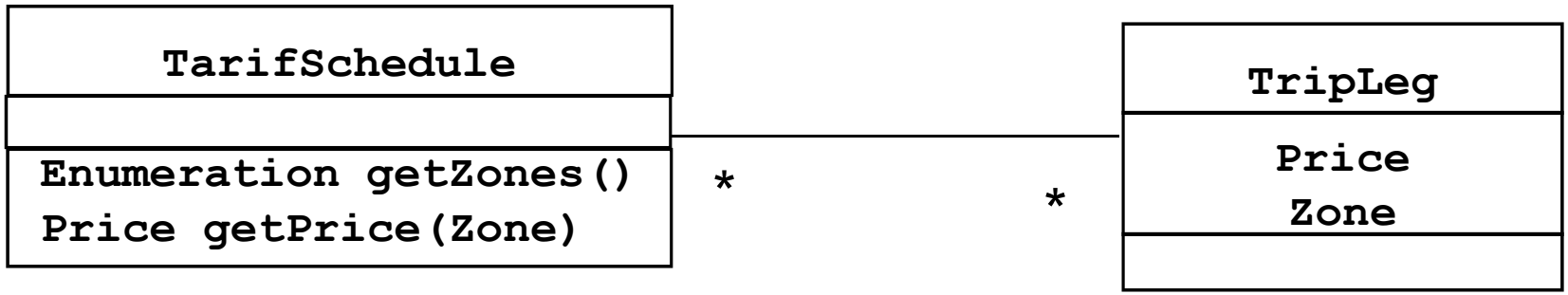
- **Actor**
  - An entity outside the system to be modeled, interacting with the system ("Passenger")
- **Class**
  - An abstraction modeling an entity in the application or solution domain
  - The class is part of the system model ("User", "Ticket distributor", "Server")
- **Object**
  - A specific instance of a class ("Joe, the passenger who is purchasing a ticket from the ticket distributor").

Instances



- An ***instance*** represents a phenomenon
- The attributes are represented with their ***values***
- The name of an instance is underlined
- The name can contain only the class name of the instance (anonymous instance)

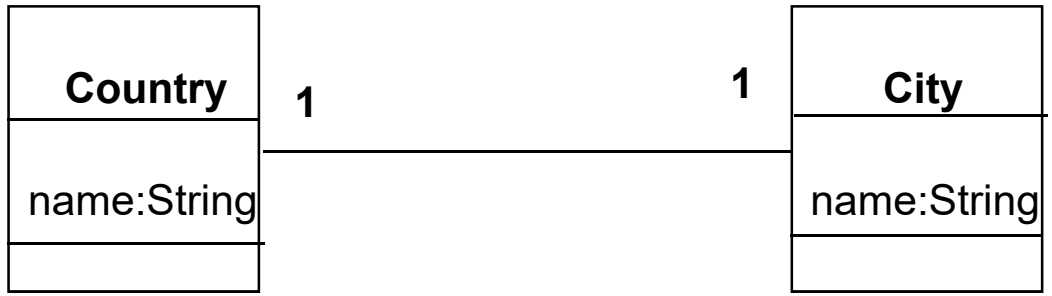
Associations



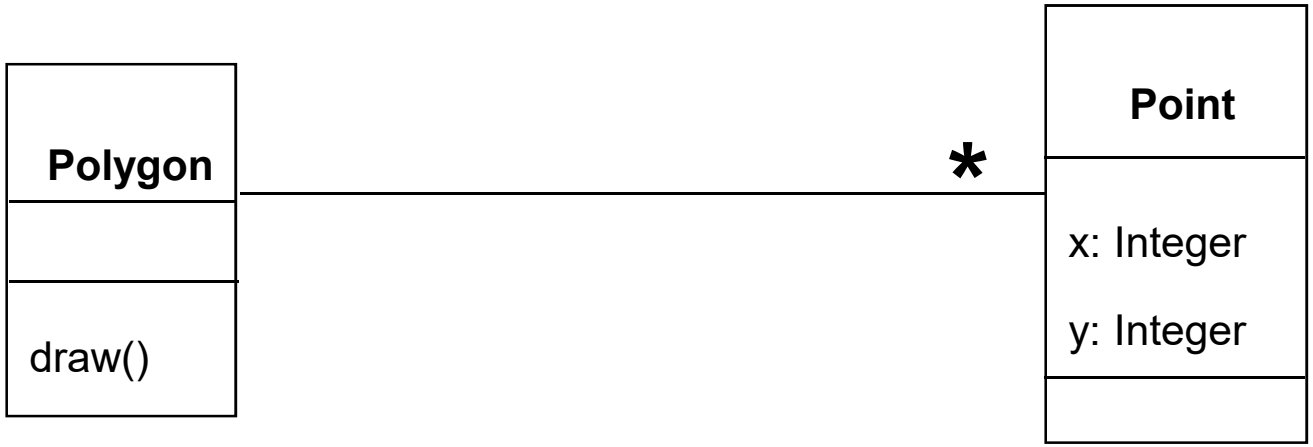
Associations denote relationships between classes

The multiplicity of an association end denotes how many objects the instance of a class can legitimately reference.

1-to-1 and 1-to-many Associations



1-to-1 association



1-to-many association

Many-to-many Associations



- A stock exchange lists many companies.
- Each company is identified by a ticker symbol

From Problem Statement To Object Model

Problem Statement: A stock exchange lists many companies.
Each company is uniquely identified by a ticker symbol

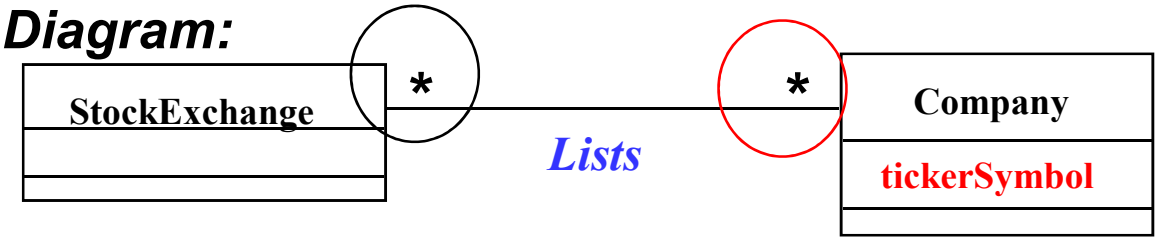
Class Diagram:



From Problem Statement to Code

Problem Statement : A stock exchange lists many companies. Each company is identified by a ticker symbol

Class Diagram:



Java Code

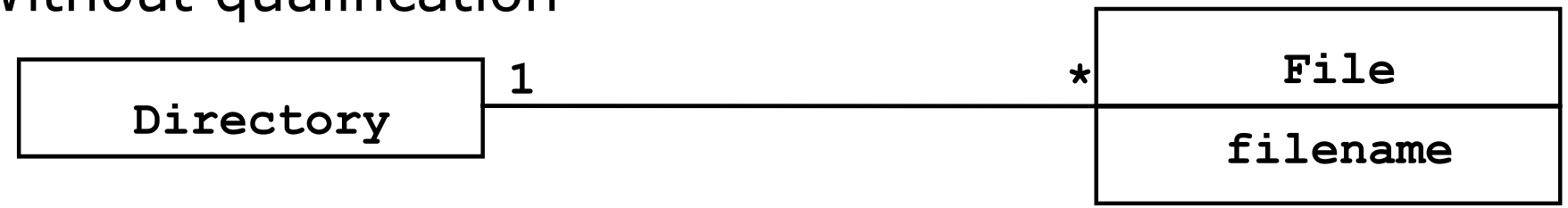
```
public class StockExchange
{
 private Vector m_Company = new Vector();
};

public class Company
{
 public int m_tickerSymbol;
 private Vector m_StockExchange = new Vector();
};
```

Associations are mapped to Attributes!

Qualifiers

Without qualification

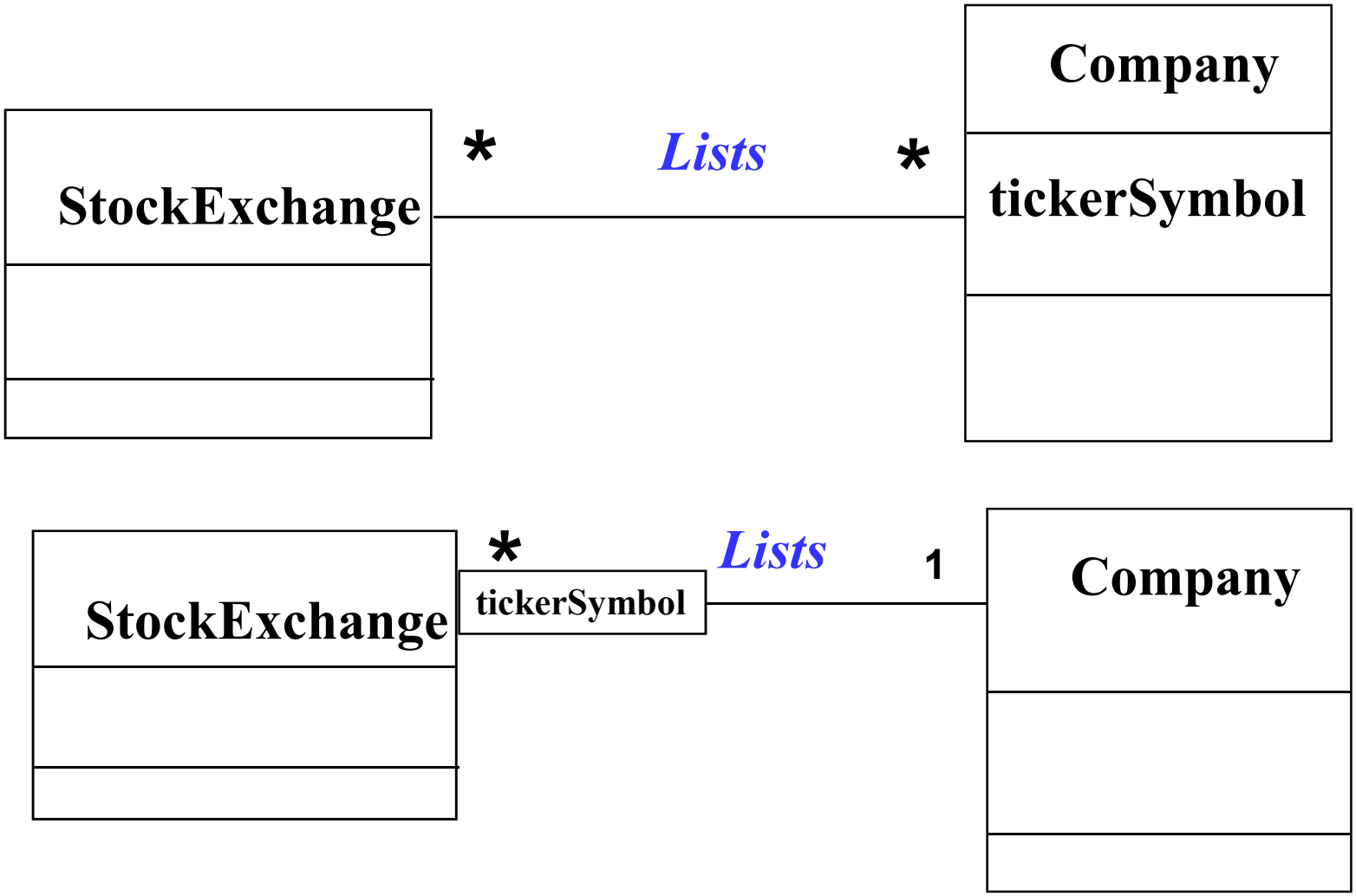


With qualification



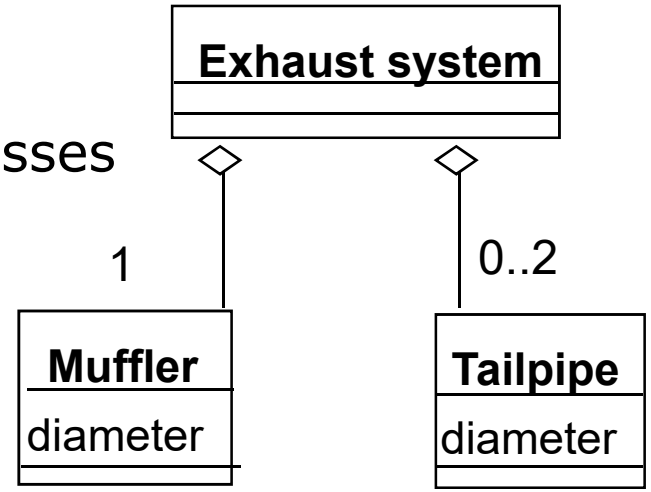
- Qualifiers can be used to reduce the multiplicity of an association

Qualification: Another Example

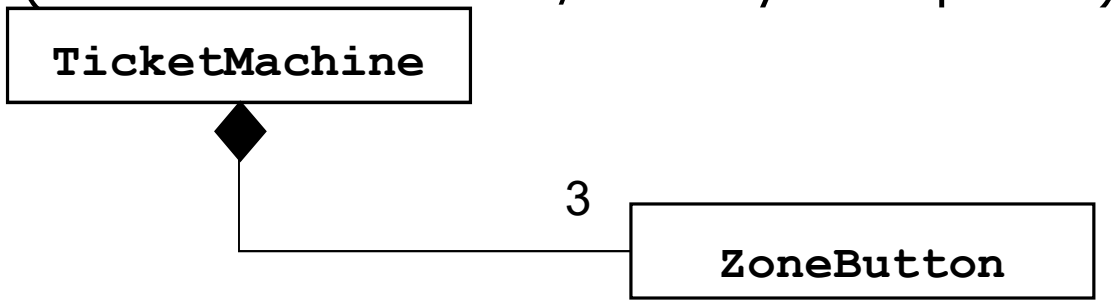


Aggregation

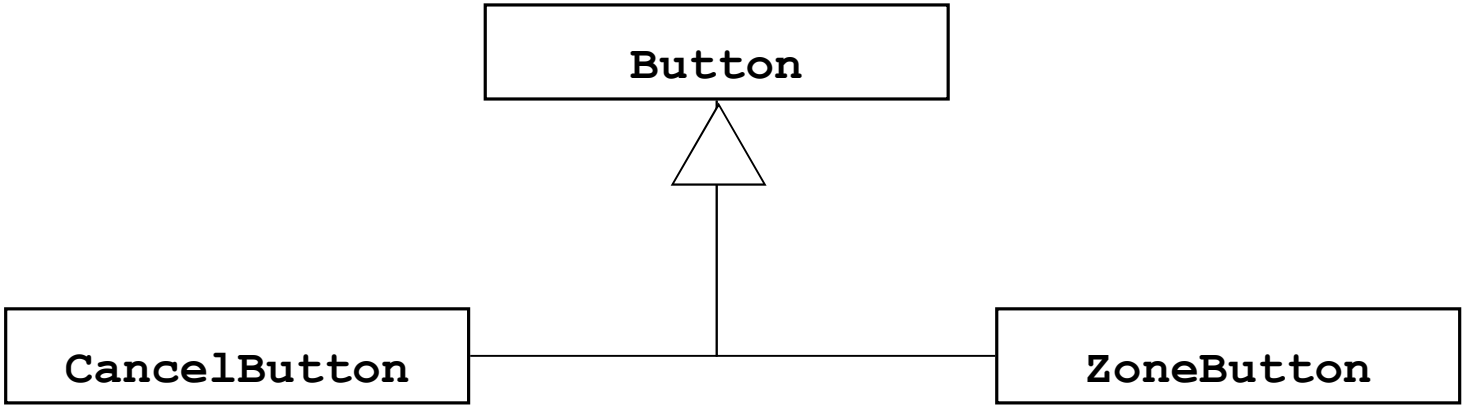
- An *aggregation* is a special case of association denoting a "consists-of" hierarchy
- The *aggregate* is the parent class, the components are the children classes



A solid diamond denotes *composition*: A strong form of aggregation where the *life time of the component instances* is controlled by the aggregate. That is, the parts don't exist on their won ("the whole controls/destroys the parts")



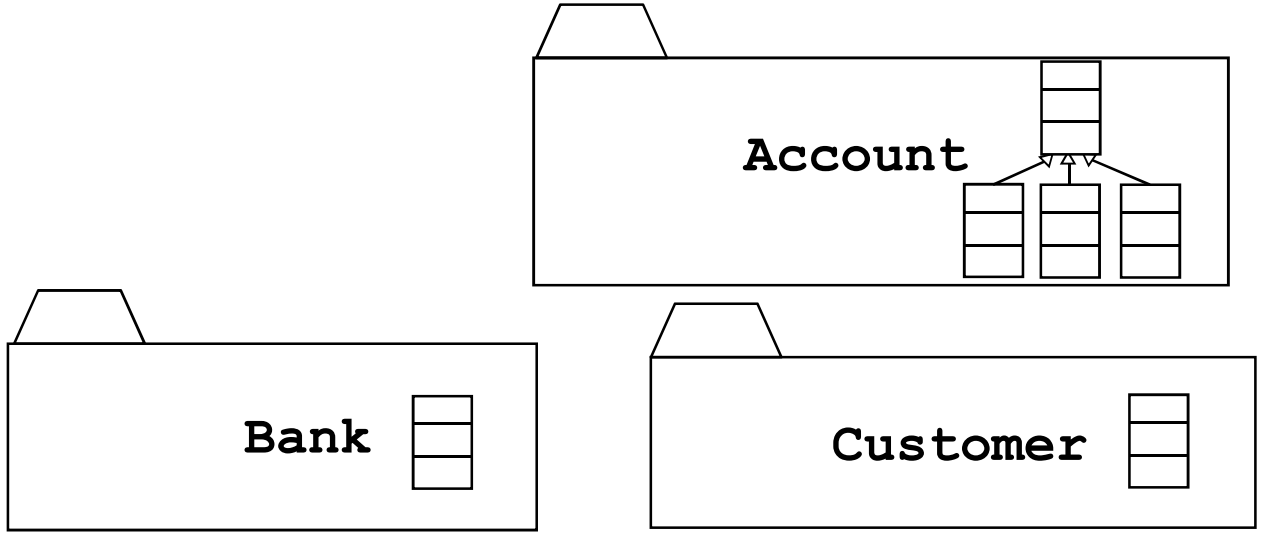
Inheritance



- *Inheritance* is another special case of an association denoting a "kind-of" hierarchy
- Inheritance simplifies the analysis model by introducing a taxonomy
- The **children classes** inherit the attributes and operations of the **parent class.**

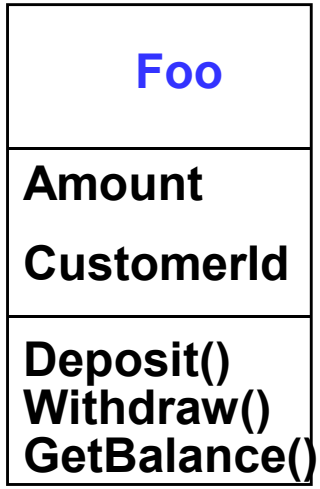
Packages

- Packages help you to organize UML models to increase their readability
- We can use the UML package mechanism to organize classes into subsystems



- Any complex system can be decomposed into subsystems, where each subsystem is modeled as a package.

Object Modeling in Practice

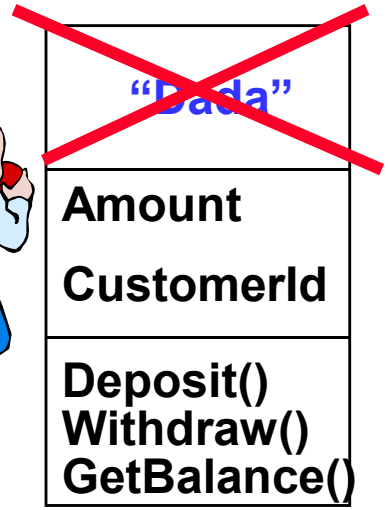


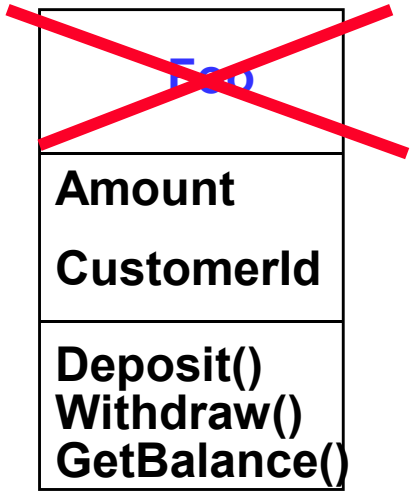
Class Identification: Name of Class, Attributes and Methods

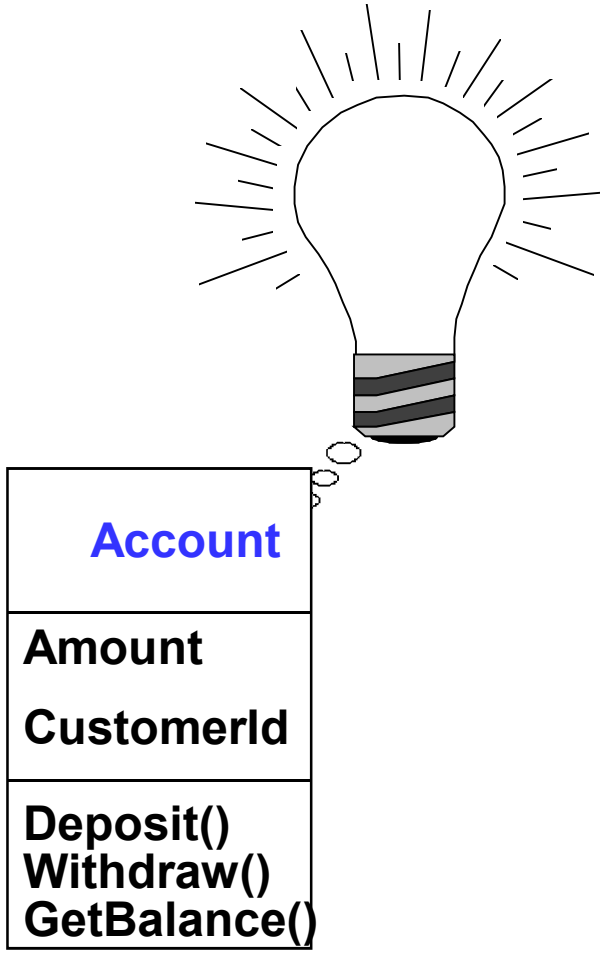
Is Foo the right name?

Object Modeling in Practice: Brainstorming



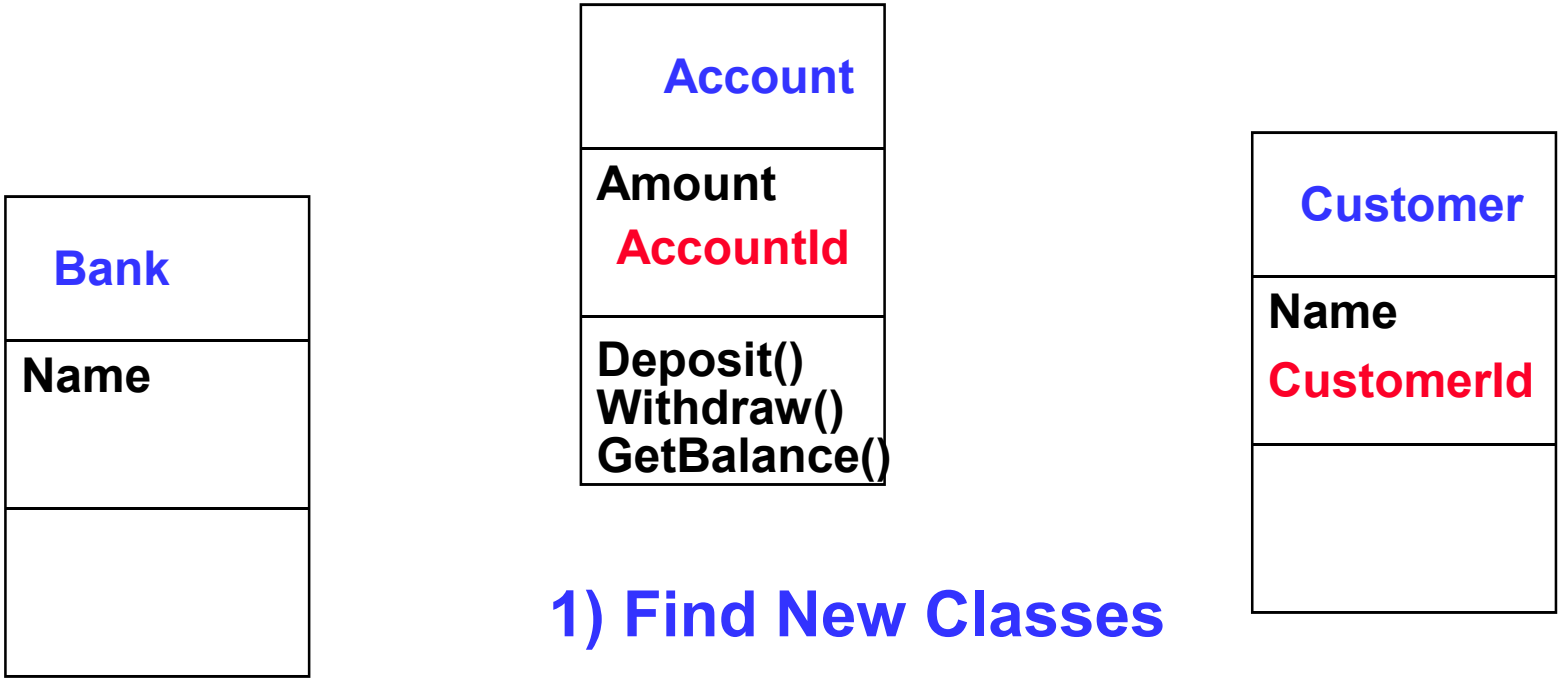






Is Foo the right name?

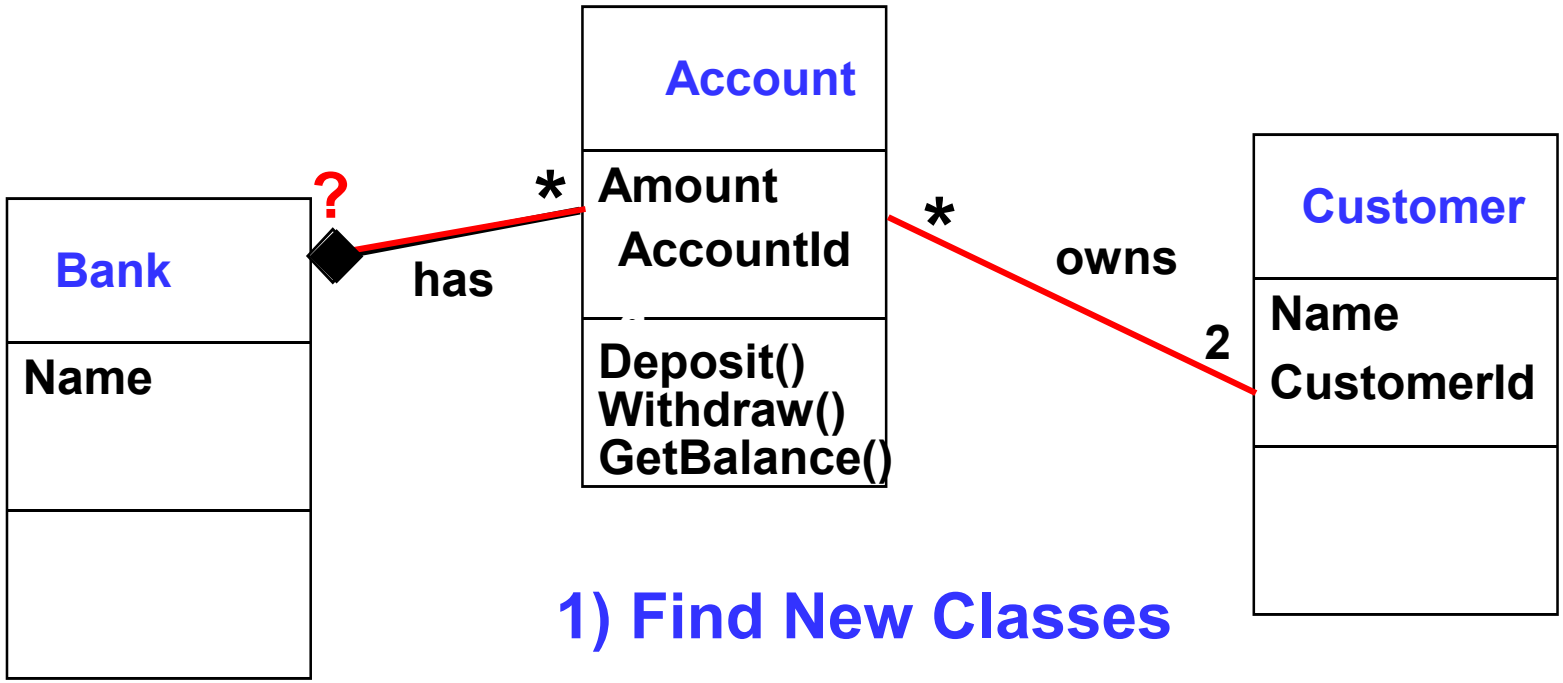
Object Modeling in Practice: More classes



1) Find New Classes

2) Review Names, Attributes and Methods

Object Modeling in Practice: Associations



1) Find New Classes

2) Review Names, Attributes and Methods

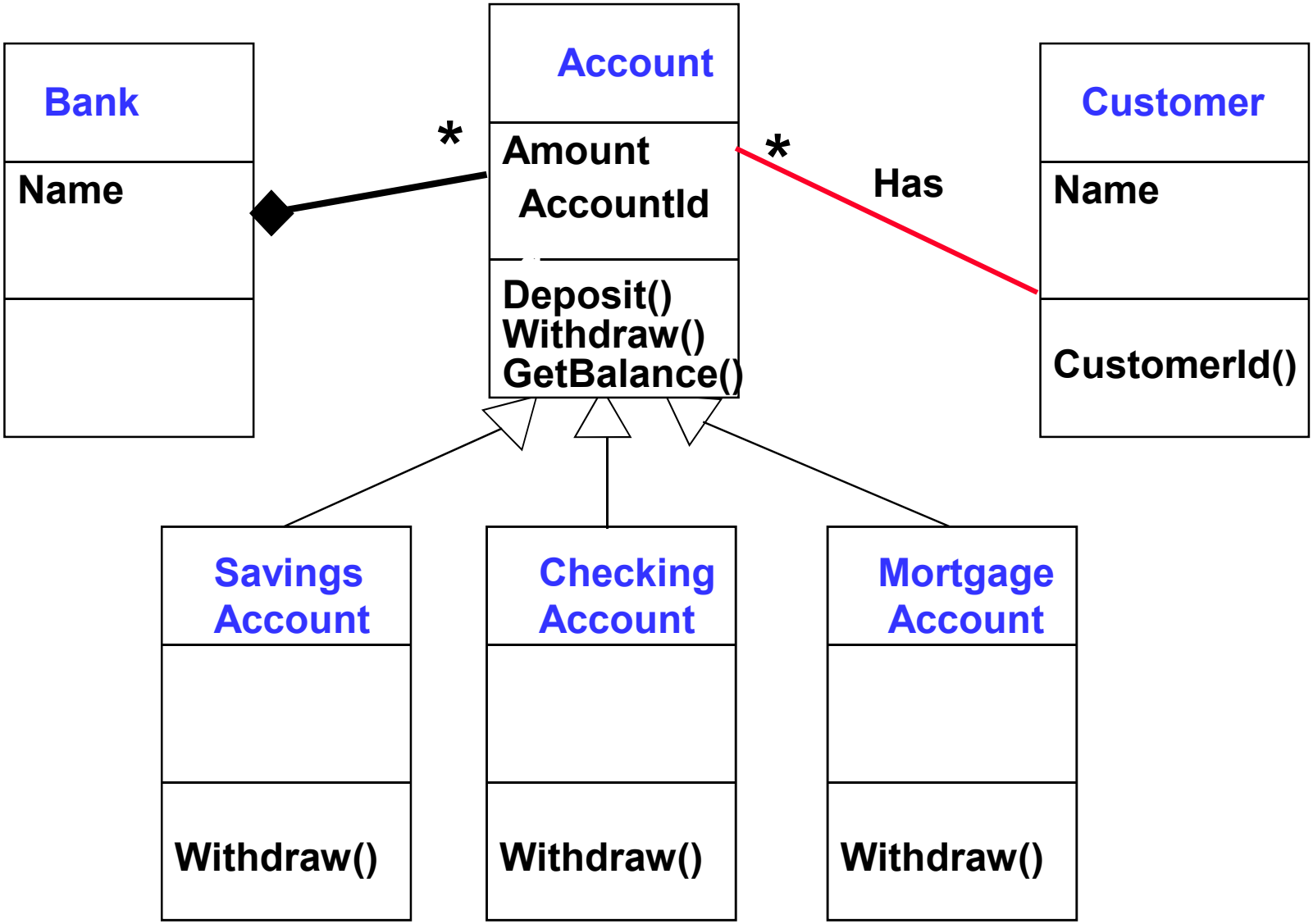
3) Find Associations between Classes

4) Label the generic assocations

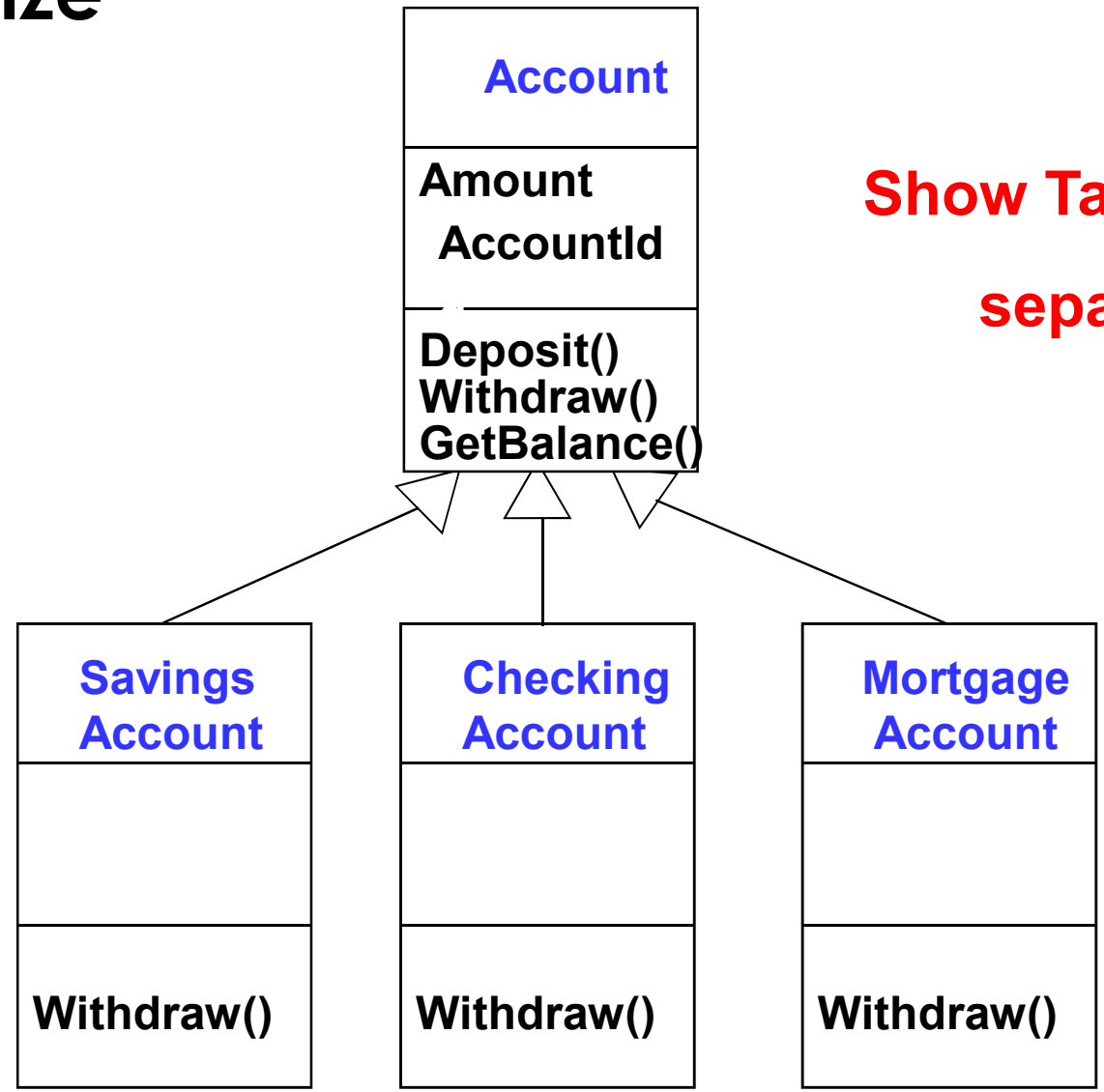
5) Determine the multiplicity of the assocations

6) Review associations

Practice Object Modeling: Find Taxonomies

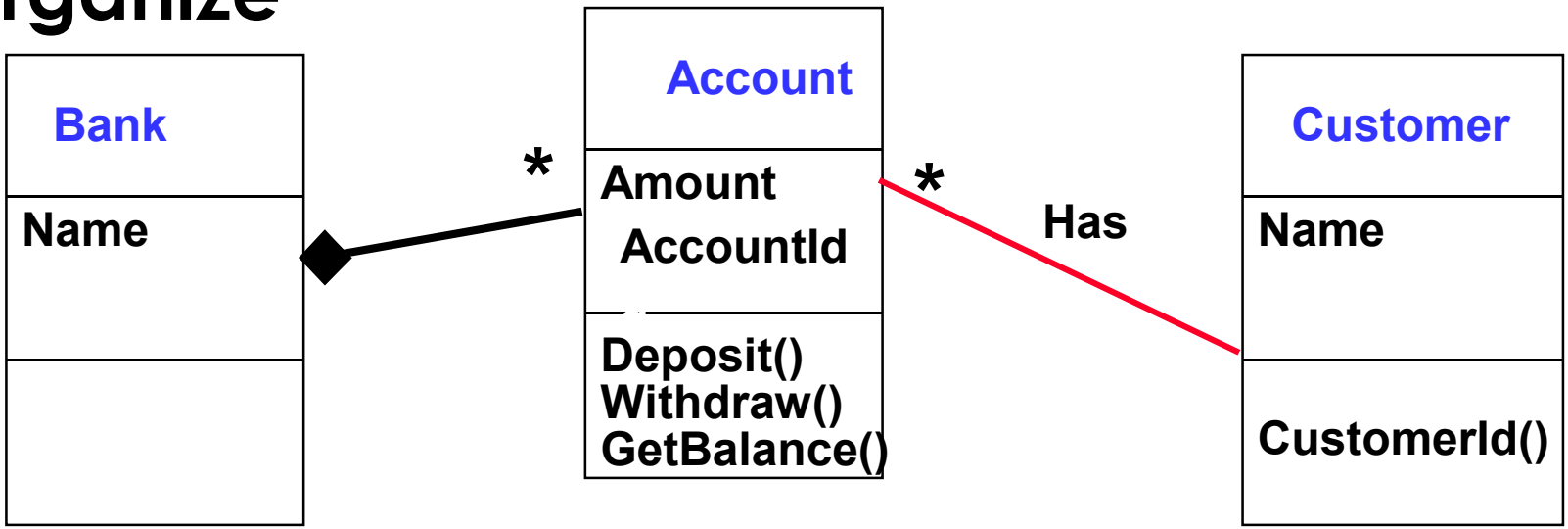


Practice Object Modeling: Simplify, Organize



Show Taxonomies separately

Practice Object Modeling: Simplify, Organize

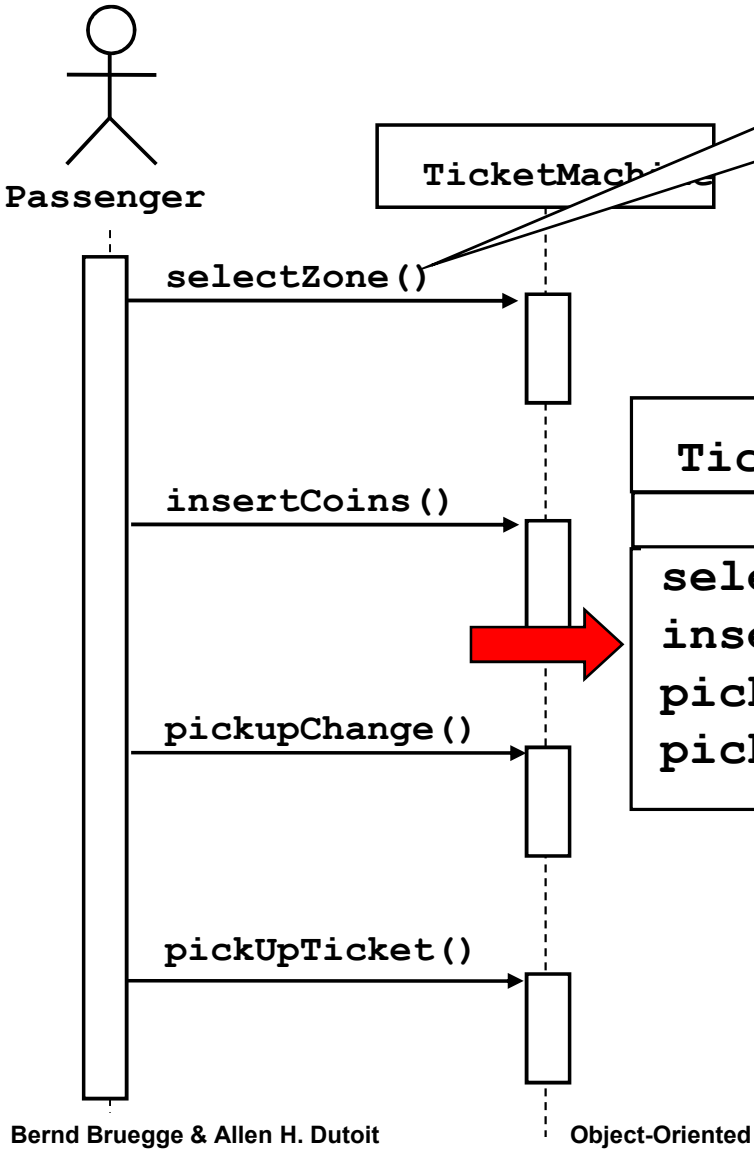


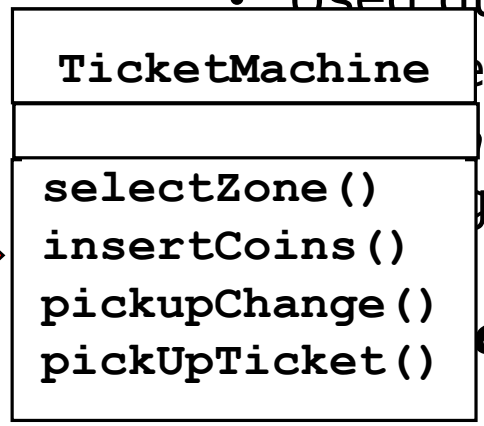
Use the 7+-2 heuristics

or better 5+-2!

Sequence Diagrams

Focus on Controlflow



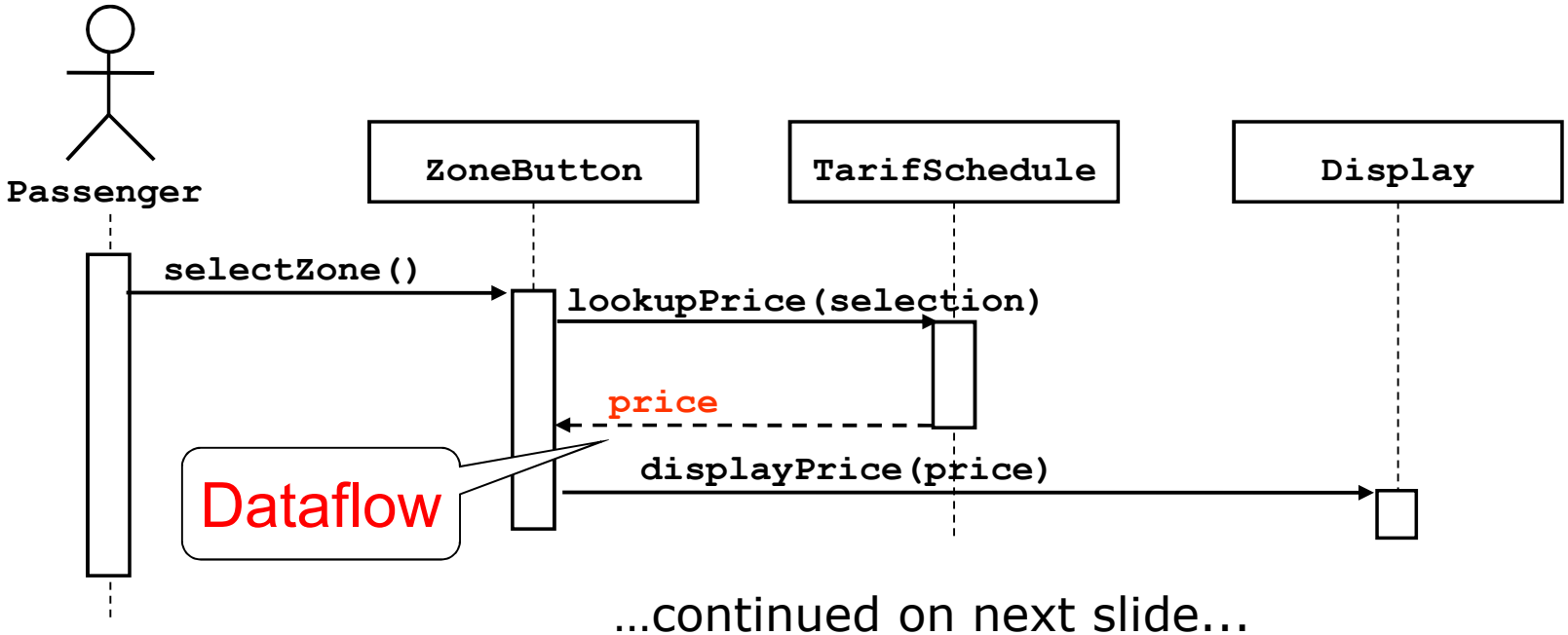


- Used during analysis
  - To refine use case descriptions
  - to find additional objects ("participating objects")
- Used during system design
  - efine ubsystem interfaces
  - ces ar
  - les. ***Act***

Messages -> Operations on participating Object

- ***es*** are represented by lines
- ***Messages*** are represented by arrows
- ***Activations*** are represented by narrow rectangles.

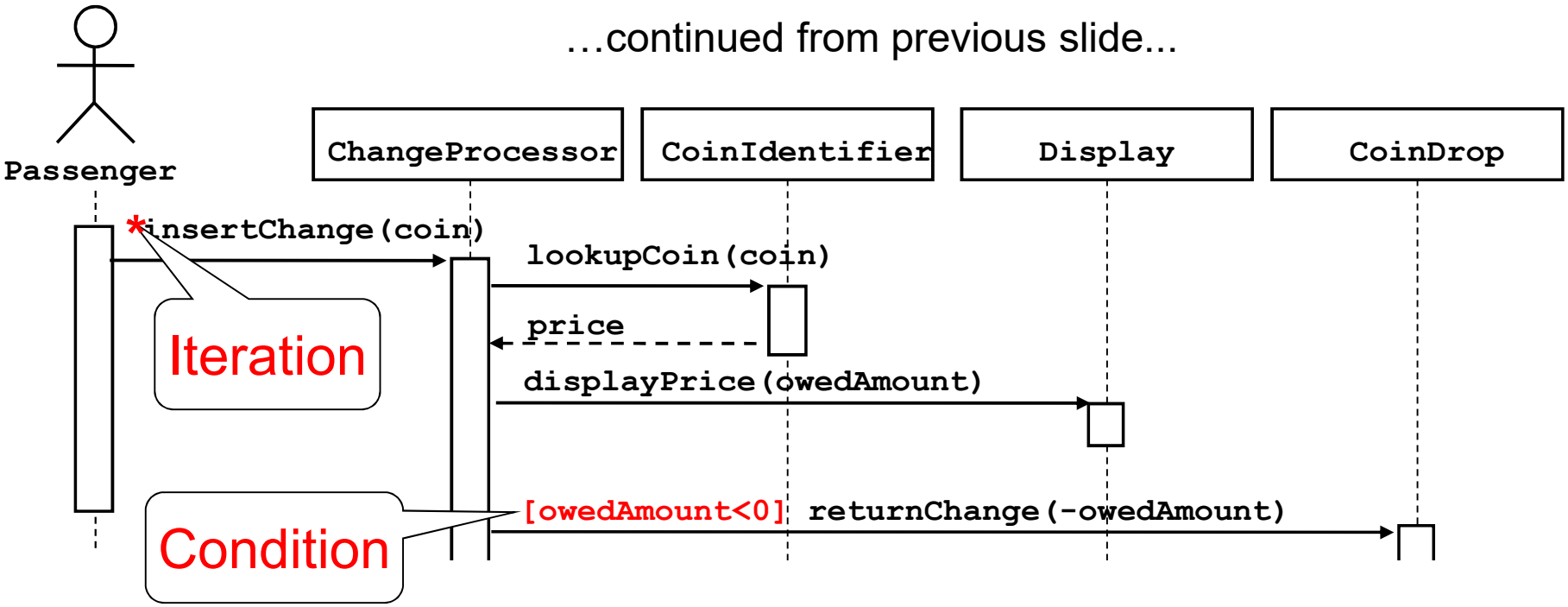
Sequence Diagrams can also model the Flow of Data



...continued on next slide...

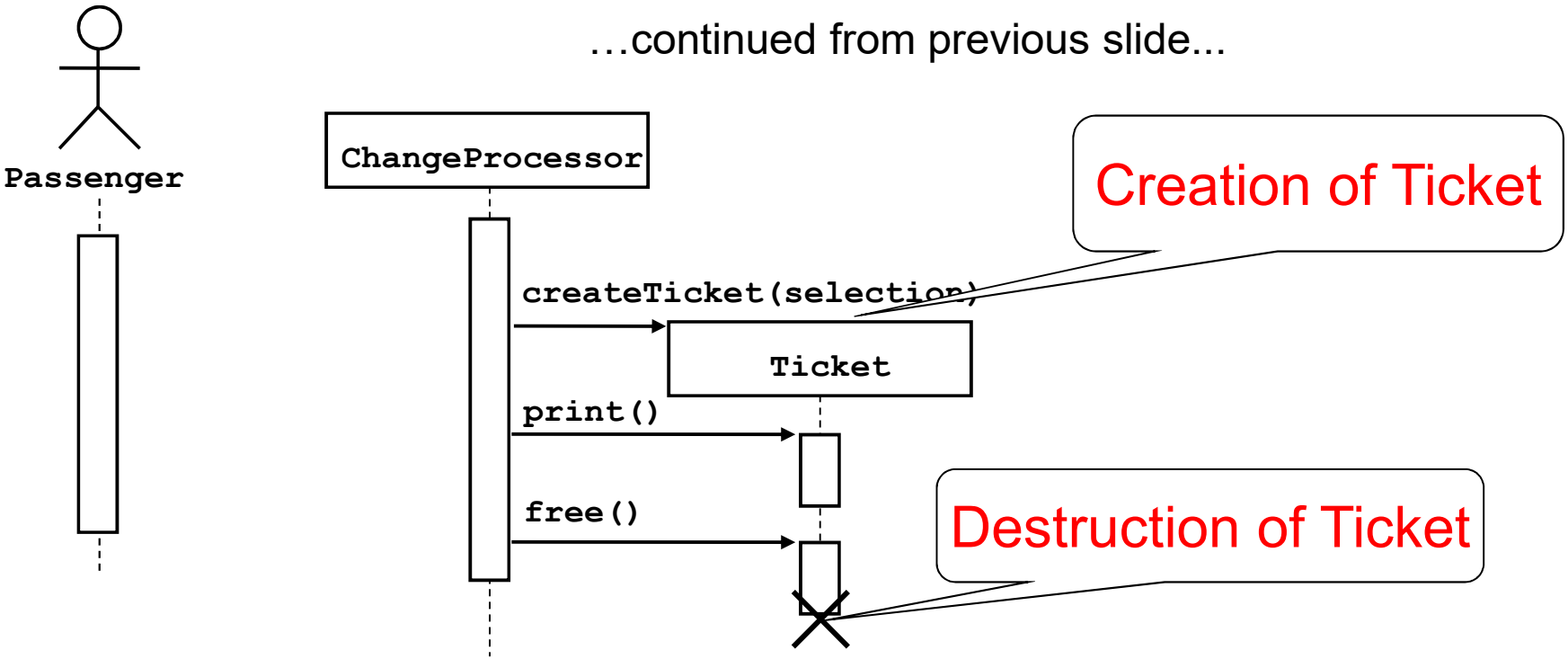
- The source of an arrow indicates the activation which sent the message
- Horizontal dashed arrows indicate data flow, for example return results from a message

Sequence Diagrams: Iteration & Condition



- Iteration is denoted by a * preceding the message name
- Condition is denoted by boolean expression in [ ] before the message name

Creation and destruction



- Creation is denoted by a message arrow pointing to the object
- Destruction is denoted by an X mark at the end of the destruction activation
  - In garbage collection environments, destruction can be used to denote the end of the useful life of an object.

Sequence Diagram Properties

- UML sequence diagram represent *behavior in terms of interactions*
- Useful to identify or find missing objects
- Time consuming to build, but worth the investment
- Complement the class diagrams (which represent structure).

Outline of this Class

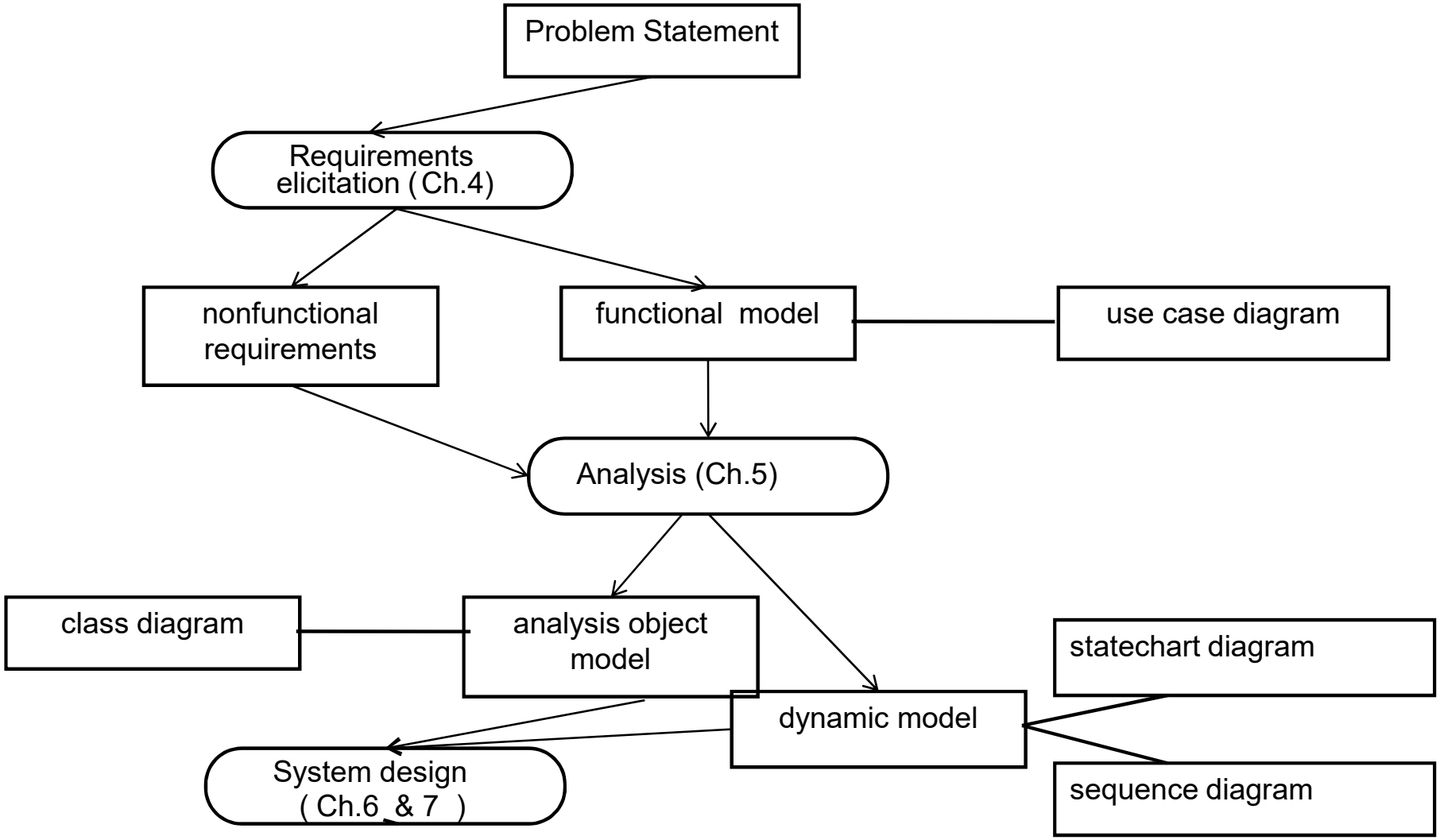
- What is UML?
- A more detailed view on
  - ✓ Use case diagrams
  - ✓ Class diagrams
  - ✓ Sequence diagrams
  - ➢ Activity diagrams

UML Activity Diagrams

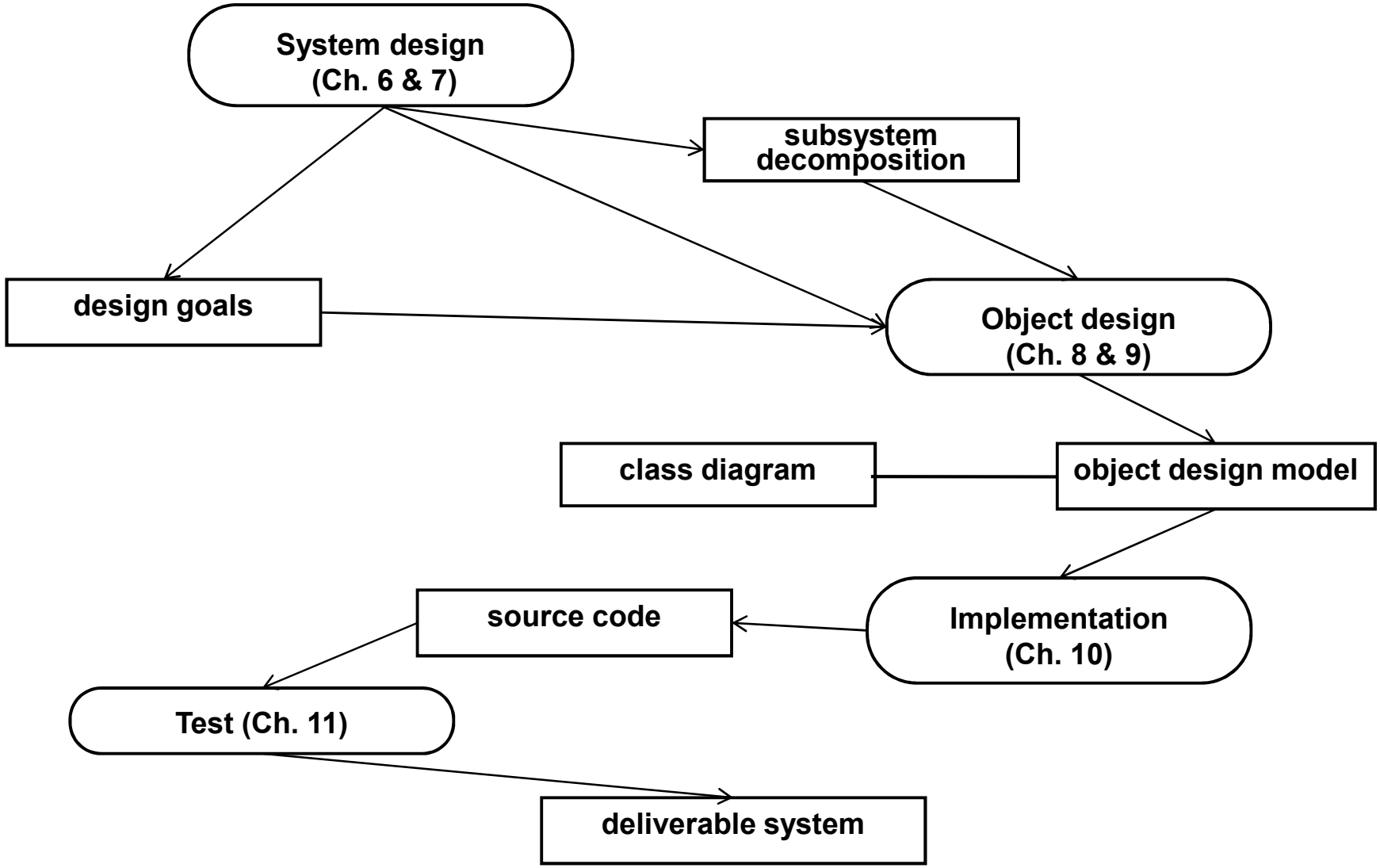
An activity diagram consists of nodes and edges

- **Nodes describe activities and objects**
  - Control nodes
  - Executable nodes
    - Most prominent: **Action**
  - Object nodes
    - E.g. a document
- **Edge** is a directed connection between nodes
  - There are two types of edges
    - Control flow edges
    - Object flow edges

Example: Structure of the Text Book

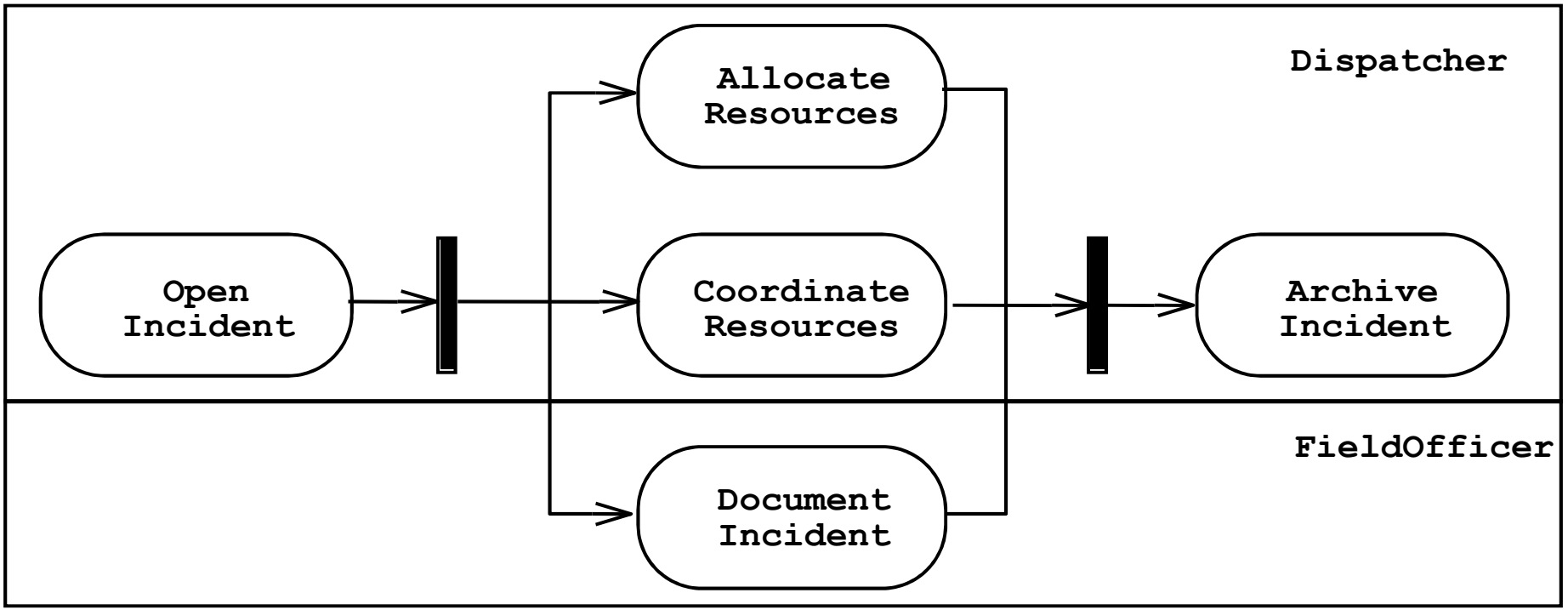


Example: Structure of the Text Book (2)



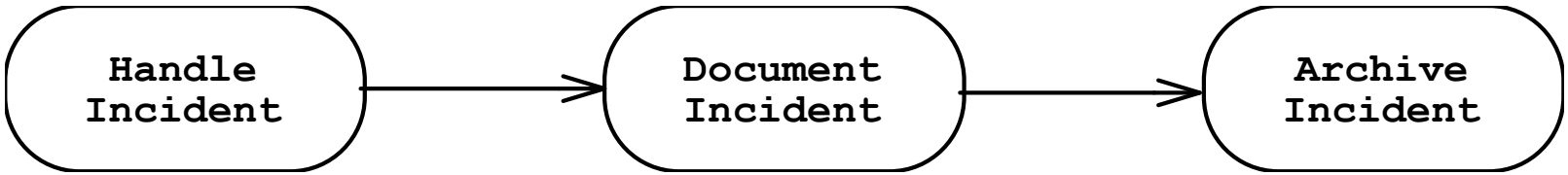
Activity Diagrams: Grouping of Activities

- Activities may be grouped into swimlanes to denote the object or subsystem that implements the activities.



State Chart Diagrams vs Activity Diagrams

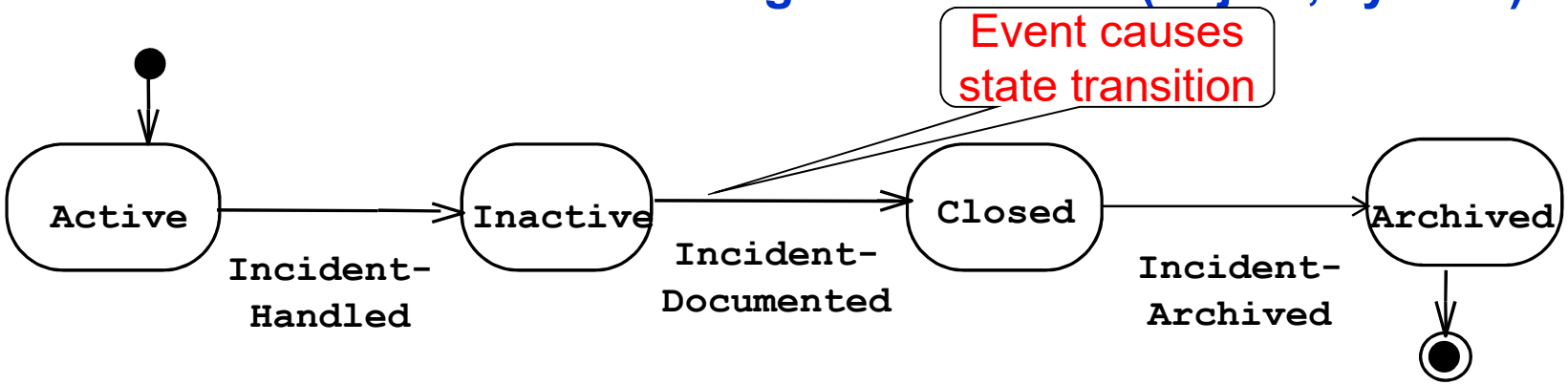
- An activity diagram that contains only activities can be seen as a special case of a state chart diagram
- Such an activity diagram is useful to describe the overall workflow of a system



Statechart Diagram vs Activity Diagram

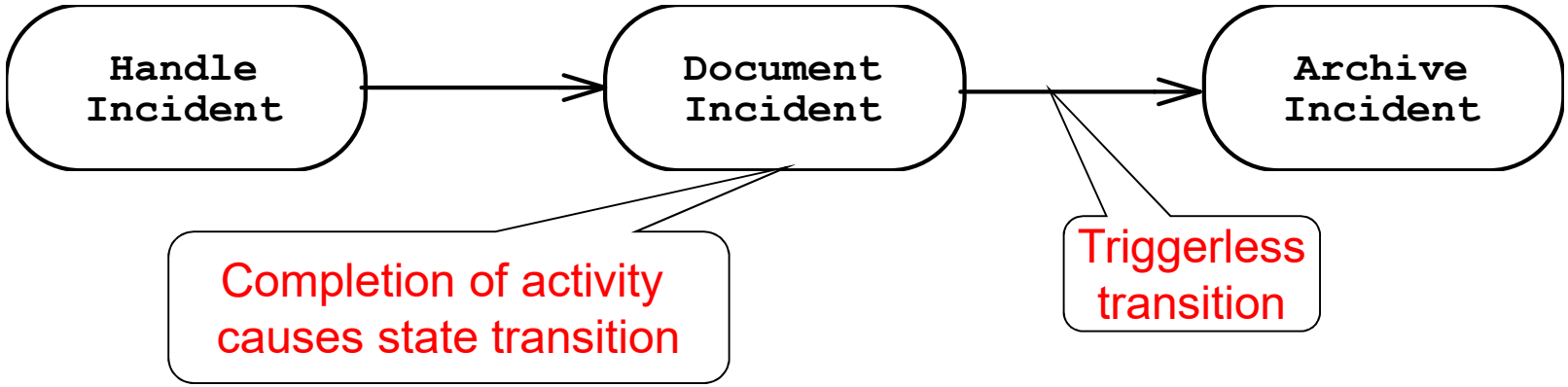
Statechart Diagram for Incident

Focus on the set of attributes of a single abstraction (object, system)



Activity Diagram for Incident

(Focus on dataflow in a system)



UML Summary

- UML provides a wide variety of notations for representing many aspects of software development
  - Powerful, but complex
- UML is a programming language
  - Can be misused to generate unreadable models
  - Can be misunderstood when using too many exotic features
- We concentrated on a few notations:
  - Functional model: Use case diagram
  - Object model: class diagram
  - Dynamic model: sequence diagrams, statechart and activity diagrams.